NAGABHUSHANA RAJU S

"Turning Data & AI into Real-World Innovation with an Entrepreneurial Mindset"-
| MAHARAJA INSTITUTE OF TECHNOLOGY MYSORE
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Professional Summary:
Passionate Data and AI/ML Engineer with an entrepreneurial mindset, skilled in building intelligent, data-driven systems and scalable solutions that deliver measurable impact. Experienced in applying machine learning to complex problems, optimizing data pipelines, and developing AI-powered applications. Driven by innovation and experimentation, I bridge engineering and entrepreneurship to transform concepts into meaningful real-world outcomes.

EDUCATION

MAHARAJA INSTITUTE OF TECHNOLOGY MYSORE — *ENGINEERING* *Oct 2022 – Present*
Mysore

CBSE (Purna Chetana Public School) — *10TH* *Jan 2018 – Apr 2019*
Mysore, India

Manasarowar Pushkarini Vidhyasrama PU College — *12TH/PUC* *Dec 2019 – Jul 2021*
Mysore, India

SKILLS

- AIML , NLP , Model -Training & Testing
- Programming & Databases : (C , HTML , Java, MongoDB, MySQL, Python , SQL,...)
- Git (VersionControll) & environment (conda/mamba) & Containerization(Docker)
- Business Intelligence & Analysis Data Analysis (Data-Visualization ,Exploratory Data Analysis ,MachineLearning ,Multimodal AI ,Recommendation System Techniques)
- Design & Professional Skills Ai tools- Computer-Aided Design (CAD) Electronic Workbench MS-Office
- Advanced Prompt Engineering
- N8n (Automation) & Latest tech
- Vibe Coding

PROJECTS

— *Expense Tracker Web App* *Mar 2025 – Apr 2025*
My Laptop

- Built a responsive and interactive Expense Tracker Web Application that allows users to manage their income and expenses, track spending habits, and visualize financial data in real time. The app helps users maintain a monthly budget and stay financially organized with a simple and intuitive interface. Key Features: Add income and expense entries with category tags and timestamps Display running balance, total income, and total expenses Visual charts (e.g., pie or bar charts) to track spending distribution Edit and delete transactions Persistent data storage using localStorage or integrated with backend/database Mobile-friendly, responsive design

— *Real-Time Sign Language Interpreter with Multi-Modal AI Perception {Facial & Gender Recognition With Emotion Analysis }* *Jun 2024 – Feb 2025*

My Laptop

- Developed a high-performance, real-time video communication platform designed to bridge the communication gap for sign language users. The system captures a webcam feed and processes it through a decoupled, multi-threaded Python backend built with FastAPI and WebSockets.
- The core of the project is a multi-modal AI pipeline that runs several concurrent modules:
- A MediaPipe Holistic model to detect and track 500+ face, hand, and pose landmarks.
- A One-Euro Filter implemented in Python to apply temporal smoothing, significantly reducing landmark jitter and stabilizing the output.
- A Deep Learning (DNN) Face Processor using pre-trained Caffe models to perform real-time gender classification.
- The results are streamed to a JavaScript frontend, which renders the processed video feed with smoothed landmarks and dynamic UI overlays, such as colored outlines around faces indicating the detected gender. The entire application is built on a modular architecture (MASE) designed for low latency and future expansion.

— *Ola Electric Clone* *Feb 2025 – Feb 2025*

My Laptop

- The Ola Electric Clone is a full-stack web application inspired by the official Ola Electric platform. It simulates a seamless user experience for browsing, booking, and managing electric scooters. The platform aims to replicate key features of Ola Electric, including vehicle listings, test ride bookings, service scheduling, and charging station locators, while also offering an admin panel for scooter and order management.

— *Multimodal AI for early disease detection* *Jun 2025 – Present*

My Laptop & GitHUB

- Multimodal AI for Early Disease Detection is an advanced AI-driven healthcare system that leverages multiple data types—such as medical images (e.g., X-rays, MRIs), electronic health records (EHRs), genetic information, and patient symptoms—to detect diseases in their early stages with higher accuracy and efficiency. Traditional diagnostic models often rely on a single source of data, which can limit the accuracy and generalizability of disease prediction. This project addresses that limitation by combining multiple data modalities using deep-learning and data fusion techniques. The integration enables the model to understand complex patterns across different data sources and provide a more comprehensive and early diagnosis.

CERTIFICATIONS

Indian Institute Of Technology + Remark Skill = Technical Workshop on Data Science — *Apr 2025 – Apr 2025*

Remark Skill Education (in association with Elan & nvision, IIT Hyderabad)

IIT Hyderabad, India

- A technical workshop on Data Science held at the Indian Institute of Technology Hyderabad as part of the Elan & nvision event.

Jul 2022 – Sep 2022

Title: Computer Applications & Programming in C — *Indian Institute of Computer Technology (IICT) – Guru Raghavendra Computer Training Center, Mysore*
Mysore, India

- Successfully completed training in Computer Concepts, MS Office (Word, Excel, PowerPoint), Internet, Nudi, and Programming in C.

Title: Hack 4 Mysore (Hackathon Participation) — *Institute/Organisation: Maharaja Institute of Technology, Mysore* *Aug 2023 – Aug 2023*
Location: Mysore, India

- Participated in Hack 4 Mysore, a competitive hackathon hosted by MIT Mysore, as a member of Team 688639-URK436L7. Developed a Web2-based fitness and wellness platform focused on women's health, specifically addressing menstrual well-being. The project featured a personalized workout planner that suggested safe exercises during menstrual cycles, highlighted movements to avoid, and included login functionality, task-setting (to-do list), and schedule planning. The solution aimed to empower women to manage pain and maintain healthy routines through technology-driven guidance.

Title:30 days : Data Analytics using Power BI (Workshop & End-to-End Project) — *TechTip24 /Skill Course E-Learning Platform* *Apr 2025 – May 2025*
Online / Mumbai, India

- Description: An e-learning micro course focused on Power BI, covering essential tools and techniques for data visualization and business intelligence over a 30-day period.

Be 10x -->Microsoft Excel Using AI Workshop — *BE 10 x || QM OfficeMaster* *Oct 2024 – Oct 2024*
Online (inferred from digital certificate and verification link)

- A workshop focused on leveraging Artificial Intelligence to master Microsoft Excel, including the use of over 200 formulas and rapid dashboard creation.

WORK EXPERIENCE

Getskilled (A unit of GyaanKool Research labs) — *Software Development Intern* *Apr 2025 – Aug 2025*
Bangalore [offline/online]

- As an Intern, I contributed to the development of a multimodal machine learning system designed to detect diseases at an early stage by analyzing diverse healthcare data such as medical images, electronic health records (EHRs), and patient-reported symptoms. Key Responsibilities: Preprocessed and cleaned multimodal datasets including chest X-ray images, structured patient data, and clinical notes. Applied CNN-based models (ResNet, EfficientNet) for image classification tasks like pneumonia and lung cancer detection. Utilized NLP techniques (TF-IDF, BERT embeddings) to extract features from unstructured EHR text for comorbidity analysis. Built fusion models to combine image and text-based features for enhanced diagnostic accuracy. Evaluated models using metrics like ROC-AUC, F1-Score, and confusion matrix to identify early-stage prediction potential. Collaborated with a multidisciplinary team of data scientists and medical advisors to align ML outcomes with clinical relevance.